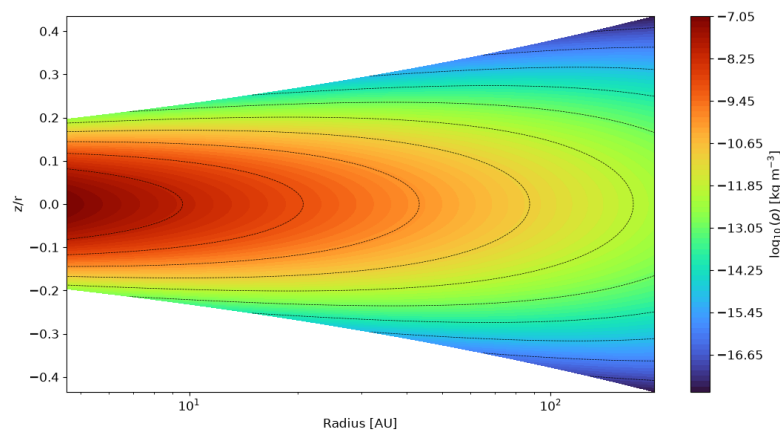


Indian Institute of Technology Indore

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Modeling the Structure of Protoplanetary Disks under simplified Astrophysical Assumptions.

Mid-Term Technical Report



Problem Statement Modeling the Structure and Evolution of Protoplanetary Disks

Difficulty Intermediate

Team Size 4 Members

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Chapter 1

Introduction

Protoplanetary disks are rotating structures of gas and dust that surround newly formed stars during the early stages of stellar evolution. These disks constitute the environments in which planets, moons, asteroids, and other planetary system bodies originate. Protoplanetary disks possess rich morphological features that include flared geometries, concentric rings, gaps, spiral arms, and dust concentrations. Understanding the physical mechanisms responsible for these structures is one of the fundamental challenges in contemporary astrophysics.

1.1 Motivation

The dynamics of a protoplanetary disk are governed by the combined effects of gravity, orbital motion, thermal pressure, and angular momentum transport. Sophisticated numerical frameworks incorporate radiative transfer, chemistry, dust evolution, and magnetohydrodynamics to model these systems with high accuracy. Although scientifically comprehensive, such models are computationally intensive and often unsuitable for educational purposes or rapid parameter exploration.

The objective of this project is to develop a lightweight and modular computational framework capable of reproducing the fundamental structure of a passive protoplanetary disk using simplified yet physically motivated analytical models. The simulator is designed to provide an intuitive platform for understanding disk physics while remaining computationally efficient and easily extensible.

1.2 Scope of the Present Work

The present implementation models a static, axisymmetric, passively irradiated protoplanetary disk under the assumption of vertical hydrostatic equilibrium. The simulator computes the radial temperature profile, surface density distribution, pressure scale height, Keplerian rotational velocity, and the resulting two-dimensional gas density structure.

Complex physical processes such as radiative transfer, chemistry, magnetic fields, dust settling, planet-disk interactions, and time-dependent viscous evolution are intentionally excluded from the baseline implementation in order to maintain computational simplicity. These features are identified as future extensions of the project.

Chapter 2

Physical Model

In the present work, rather than solving the complete set of magnetohydrodynamic and radiative transfer equations, the simulator employs a simplified analytical description that captures the dominant physical processes governing the disk morphology.

The disk is represented in a cylindrical coordinate system (R, ϕ, z) , where R denotes the radial distance from the central star, ϕ represents the azimuthal angle, and z is the height above the disk midplane. Owing to the approximately axisymmetric nature of most observed protoplanetary disks, variations along the azimuthal direction are neglected, reducing the problem to a two-dimensional (R, z) description.

2.1 Physical Assumptions

A number of simplifying assumptions commonly adopted in classical disk theory are employed in the project. These assumptions are physically motivated and are appropriate for modelling the large-scale morphology of passive protoplanetary disks.

Geometrically Thin Disk

The disk is assumed to be geometrically thin such that its vertical extent is much smaller than its radial extent,

$$H \ll R, \tag{2.1}$$

where H is the pressure scale height and R is the cylindrical radius. This approximation allows the radial and vertical structures of the disk to be treated independently and is valid for the majority of observed T-Tauri and Herbig disks.

Axisymmetric Structure

The disk is assumed to possess rotational symmetry about the central star. Consequently, all physical quantities are considered independent of the azimuthal coordinate (ϕ) and depend only on the radial distance and height above the midplane,

$$\rho = \rho(R, z). \tag{2.2}$$

This assumption neglects spiral arms, vortices, warps and other localized asymmetries.

Static Disk Approximation

The present model represents a static snapshot of the disk. Time-dependent processes such as viscous evolution, accretion, photoevaporation and disk dispersal are neglected. This approximation allows the focus to remain on the equilibrium structure of the disk.

Keplerian Rotation

The gravitational potential is assumed to be dominated by the central star, while the self-gravity of the disk is neglected. Under this assumption, the gas follows Keplerian rotation throughout the disk,

$$\Omega(R) = \sqrt{\frac{GM_*}{R^3}}. \quad (2.3)$$

This is a valid approximation for most protoplanetary disks whose masses are significantly smaller than the stellar mass.

Vertical Hydrostatic Equilibrium

The vertical structure of the disk is obtained by assuming hydrostatic equilibrium. At every radius, the upward pressure gradient balances the vertical component of stellar gravity. This assumption naturally determines the pressure scale height and leads to the Gaussian density profile adopted in the present work.

Passive Irradiation

The disk is assumed to be heated primarily by stellar irradiation. Heating due to viscous accretion is neglected, making the disk a passive flared disk. Consequently, the temperature distribution is prescribed using a radial power-law profile rather than being obtained from detailed radiative transfer calculations.

Well-Mixed Gas and Dust

Gas and dust are assumed to remain perfectly coupled throughout the disk with a constant dust-to-gas mass ratio. Processes such as grain growth, radial drift, fragmentation and dust settling are beyond the scope of the current implementation.

Neglected Physical Processes

To maintain computational simplicity, several advanced physical mechanisms are intentionally omitted from the baseline model. These include detailed radiative transfer, chemical evolution, magnetic fields, non-LTE effects, disk self-gravity and planet–disk interactions. Although these processes can significantly influence disk evolution, they require substantially more sophisticated numerical treatments and are identified as potential extensions for future versions of the simulator.

Together, these assumptions provide a balance between physical realism and computational efficiency, enabling the construction of a self-consistent first-order model capable of reproducing the principal structural characteristics of passive protoplanetary disks while maintaining a modular and extensible computational framework.

Chapter 3

Mathematical Model

The objective of the present work is to reproduce the large-scale structure of a passive protoplanetary disk using a computationally efficient analytical model. Rather than solving the complete system of magnetohydrodynamic and radiative transfer equations, the simulator employs a set of simplified equations that describe the dominant physical processes governing the radial and vertical structure of the disk.

The mathematical framework is designed to satisfy three primary objectives:

- Represent the radial distribution of matter within the disk.
- Model the thermal and dynamical structure using physically motivated relations.
- Reconstruct the two-dimensional density distribution under the assumption of hydrostatic equilibrium.

Each equation implemented in the simulator corresponds to a distinct physical process and collectively forms a self-consistent model of a passive, flared protoplanetary disk.

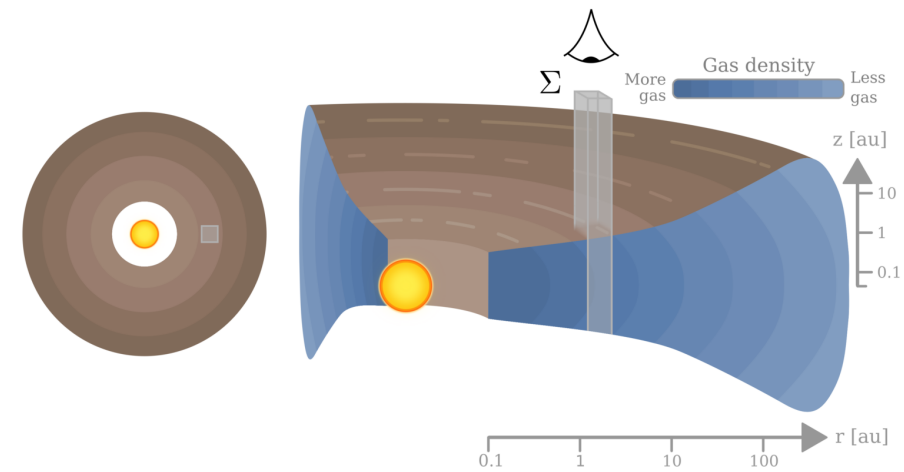


Figure 3.1: Illustration of the Disk Model

3.1 Surface Density Distribution

The surface density, denoted by $\Sigma(R)$, represents the vertically integrated mass per unit area of the disk. It determines how material is distributed with increasing distance from the central star and directly influences the total disk mass, pressure structure, and vertical density distribution.

Instead of numerically solving the viscous diffusion equation governing disk evolution, the present work adopts a widely used analytical self-similar surface density profile. This formulation naturally produces a power-law distribution in the inner disk while introducing an exponential taper towards the outer regions, closely matching observations of many protoplanetary disks.

The adopted surface density profile is

$$\Sigma(R) = \Sigma_c \left(\frac{R}{R_c} \right)^{-\gamma} \exp \left[- \left(\frac{R}{R_c} \right)^{2-\gamma} \right], \quad (3.1)$$

where

- Σ_c is the normalization constant,
- R_c is the characteristic radius,
- γ controls the radial density slope.

This equation forms the foundation for determining the total mass distribution throughout the disk.

3.2 Temperature Profile

The thermal structure of a protoplanetary disk governs several important physical quantities including the sound speed, gas pressure, and disk thickness. Since the present model considers a passive irradiated disk, the temperature is prescribed rather than obtained through detailed radiative transfer calculations.

The radial temperature profile is approximated using a simple power-law,

$$T(R) = T_0 \left(\frac{R}{R_0} \right)^{-q}, \quad (3.2)$$

where

- T_0 is the reference temperature,
- R_0 is the reference radius,
- q is the temperature exponent.

This relation captures the gradual cooling of the disk with increasing distance from the star and provides the thermal input required for subsequent calculations.

3.3 Keplerian Rotation

The orbital motion of the gas is assumed to be entirely dominated by the gravitational attraction of the central star. Under this approximation, pressure support and disk self-gravity contribute negligibly to the orbital dynamics.

The angular velocity therefore follows the Keplerian relation

$$\Omega(R) = \sqrt{\frac{GM_*}{R^3}}, \quad (3.3)$$

where

- G is the gravitational constant,
- M_* is the stellar mass,
- R is the cylindrical radius.

The corresponding orbital velocity is obtained as

$$v_\phi(R) = R\Omega(R) = \sqrt{\frac{GM_*}{R}}. \quad (3.4)$$

The velocity profile determines the dynamical behaviour of the gas and serves as an important diagnostic when comparing theoretical models with observations.

3.4 Pressure Scale Height

The vertical thickness of the disk is governed by the balance between thermal pressure and stellar gravity. This characteristic thickness is quantified by the pressure scale height.

The first step is to determine the isothermal sound speed,

$$c_s = \sqrt{\frac{k_B T}{\mu m_H}}, \quad (3.5)$$

where

- k_B is the Boltzmann constant,
- μ is the mean molecular weight,
- m_H is the hydrogen mass.

The pressure scale height then follows directly from the condition of hydrostatic equilibrium,

$$H(R) = \frac{c_s}{\Omega}. \quad (3.6)$$

The radial increase in scale height naturally produces the flared geometry commonly observed in passive protoplanetary disks.

3.5 Vertical Density Distribution

The three-dimensional density structure of the disk is reconstructed by assuming vertical hydrostatic equilibrium together with an isothermal vertical temperature profile.

Under these assumptions, the gas density follows a Gaussian distribution,

$$\rho(R, z) = \frac{\Sigma(R)}{\sqrt{2\pi}H(R)} \exp\left(-\frac{z^2}{2H(R)^2}\right). \quad (3.7)$$

The Gaussian profile concentrates the majority of the gas close to the disk midplane while allowing the density to decrease smoothly with height. This equation is used to generate the two-dimensional density maps and vertical cross-sectional visualizations produced by the simulator.

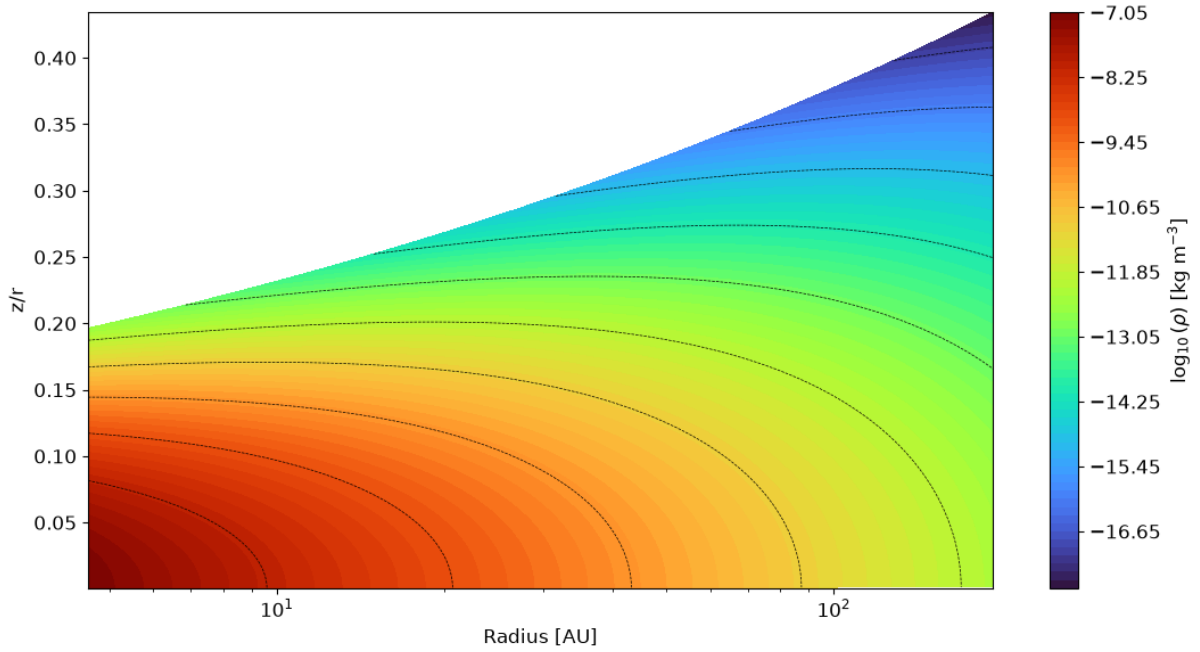


Figure 3.2: 2D-Density Distribution plot from the Model for TWHya

3.6 Summary of the Adopted Mathematical Model

The complete computational model is constructed by combining the analytical relations presented above. Each equation describes a different aspect of the disk structure while collectively producing a physically consistent representation of a passive flared disk.

Table 3.1: Summary of the equations implemented in the simulator.

Physical Quantity	Purpose
Surface Density	Radial mass distribution of the disk.
Temperature Profile	Determines the thermal structure.
Keplerian Rotation	Computes the orbital motion of the gas.
Sound Speed	Calculates thermal pressure support.
Pressure Scale Height	Determines the vertical thickness of the disk.
Gaussian Density Profile	Reconstructs the two-dimensional density structure.

Together, these equations form the mathematical foundation of the computational framework developed in this project. Their modular implementation enables individual components to be modified or replaced independently, thereby providing a flexible platform for future extensions including dust evolution, planet–disk interactions, radiative transfer, and time-dependent viscous evolution.

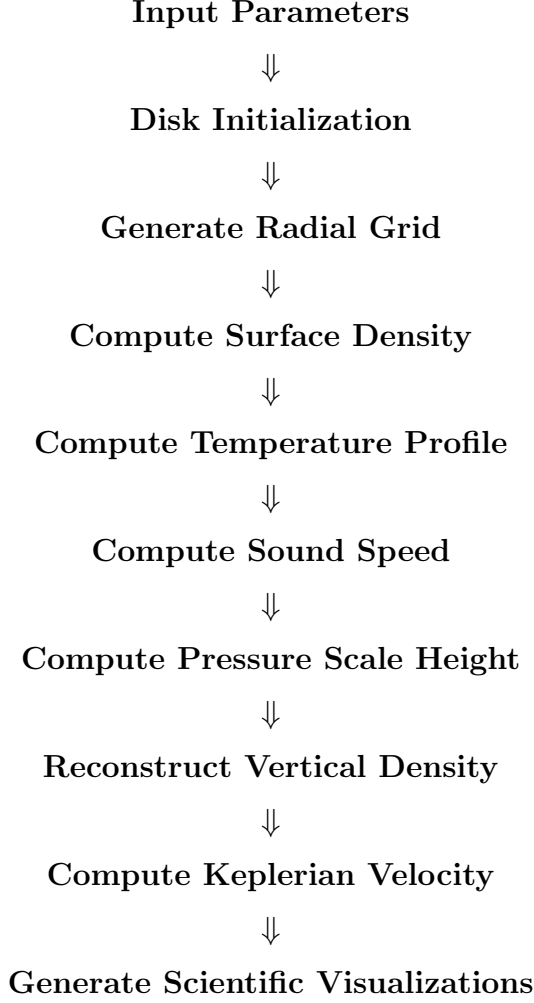
4 Numerical Implementation

The mathematical framework presented in the previous chapter is implemented as a modular Python-based simulation pipeline. Rather than treating the simulation as a single monolithic program, the implementation separates parameter initialization, physical calculations, visualization, and parameter exploration into independent components. This modular structure improves readability, simplifies debugging, and allows additional physical processes to be incorporated with minimal modifications to the existing codebase.

The simulator has been developed entirely using open-source scientific Python libraries. Numerical calculations are performed using NumPy, while visualization is carried out using Matplotlib. The modular implementation also facilitates systematic parameter studies by allowing multiple disk models to be generated and compared using a common computational framework.

4.1 Overall Simulation Pipeline

The simulator follows a sequential computational workflow beginning with user-defined stellar and disk parameters and ending with scientific visualizations of the computed disk structure.



Each stage of the pipeline uses the quantities computed in the previous stage, ensuring that all physical properties remain self-consistent throughout the simulation.

4.2 Software Architecture

The implementation is organized into independent Python classes and utility functions, each responsible for a specific stage of the simulation. This separation of responsibilities makes the code easier to maintain and enables future extensions without altering the overall workflow.

4.3 Input Parameters

The simulation requires a set of stellar and disk parameters that completely define the physical configuration of the model. These parameters include the stellar properties together with the quantities governing the radial and vertical structure of the disk.

The user specifies quantities such as the stellar mass, stellar luminosity, stellar radius, effective temperature, inner and outer disk radii, characteristic radius, total disk mass, temperature normalization, surface density exponent, temperature exponent, and mean molecular weight. In addition, the numerical resolution of the radial and vertical grids

Table 4.1: Major software components of the simulator.

Component	Responsibility
<code>default_parameters()</code>	Defines the default stellar and disk parameters used by the simulator.
<code>DiskModel</code>	Performs all physical calculations including surface density, temperature, scale height, density distribution and rotational velocity.
<code>DiskVisualizer</code>	Generates plots for the computed physical quantities including radial profiles and two-dimensional density maps.
<code>DiskComparisonVisualizer</code>	Produces comparative plots for multiple disk models generated during parameter studies.
<code>parameter_sweep()</code>	Automatically generates multiple disk models by varying a selected physical parameter while keeping the remaining parameters fixed.

can be adjusted according to the desired balance between computational cost and spatial accuracy.

For the present study, parameter values corresponding to the DIANA reference model for the T-Tauri star TW Hya are adopted as the baseline configuration. Subsequent simulations are generated by varying individual parameters while keeping the remaining quantities unchanged, allowing the physical influence of each parameter to be investigated independently.

4.4 Simulation Setup

The simulator is designed to model a generic protoplanetary disk and is therefore independent of any particular stellar system. Before each simulation, the user specifies a complete set of stellar, disk, and numerical parameters that define the physical configuration of the model. These parameters serve as the input to the computational pipeline described in the previous sections.

The stellar parameters include the stellar mass, radius, luminosity and effective temperature. The disk is characterized by quantities such as the inner and outer radii, characteristic radius, total disk mass, reference temperature, surface density exponent, temperature exponent and mean molecular weight. In addition, the numerical resolution of the radial and vertical grids is selected according to the desired balance between computational accuracy and execution time.

For validation purposes, one may initialize the simulator using parameter sets available in published disk models, such as the DIANA reference models or observationally constrained systems like TW Hya. However, the computational framework itself remains completely generic and is capable of simulating any protoplanetary disk by simply modifying the input parameters.

Table 4.2 summarizes the principal parameters required for initializing the simulation.

Table 4.2: Principal input parameters used to initialize the simulator.

Parameter	Symbol	Unit
Stellar Mass	M_*	$M_\odot$
Stellar Radius	R_*	$R_\odot$
Stellar Luminosity	L_*	$L_\odot$
Effective Temperature	T_{eff}	K
Disk Mass	M_{disk}	$M_\odot$
Inner Radius	R_{in}	AU
Outer Radius	R_{out}	AU
Characteristic Radius	R_c	AU
Reference Temperature	T_0	K
Temperature Exponent	q	—
Surface Density Exponent	γ	—
Mean Molecular Weight	μ	—
Radial Grid Size	N_R	—
Vertical Grid Size	N_Z	—

5 Results and visualizations

5.1 Visualization Capabilities

The simulator includes an integrated visualization pipeline for analysing both the radial and vertical structure of the disk.

Representative outputs generated by the implementation include

- Surface density profiles.
- Temperature profiles.
- Pressure scale height variation.
- Keplerian velocity curves.
- Midplane density profiles.
- Two-dimensional density contour maps.

- Logarithmic density distributions.
- Comparative plots for parameter variation studies.

These visualizations enable rapid inspection of the computed physical quantities and provide qualitative validation of the implemented mathematical model.

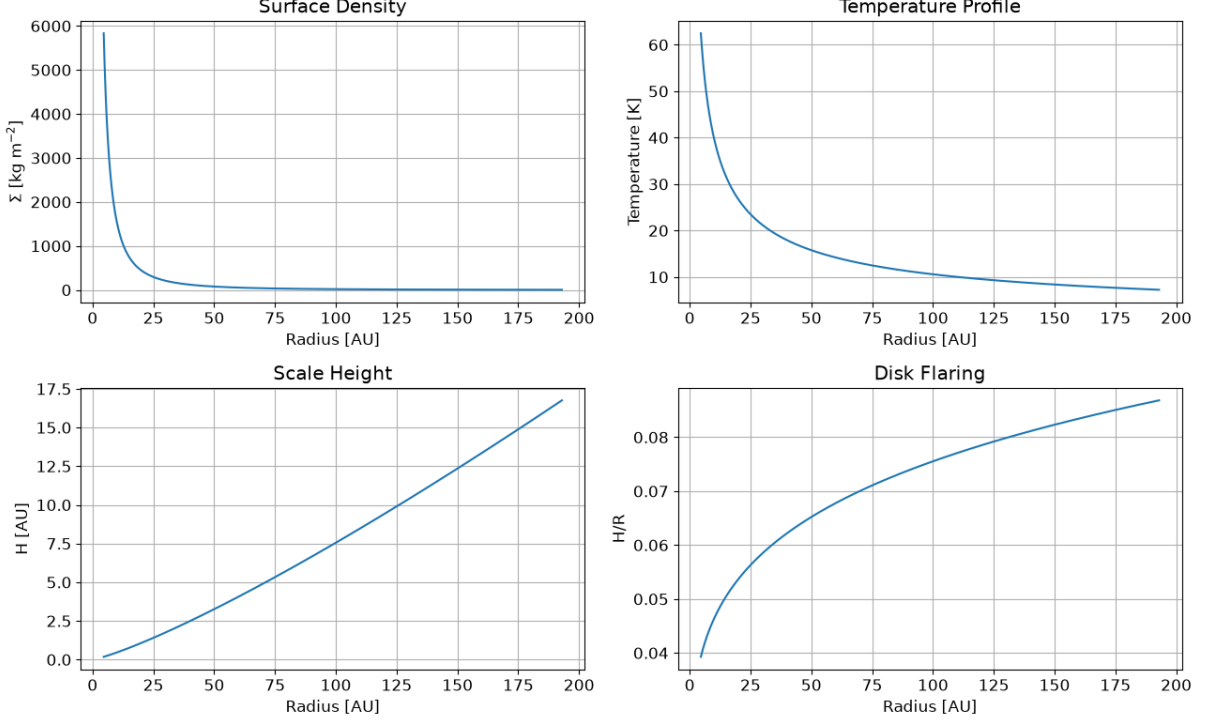


Figure 5.1: Plots from the Model for TWHya

5.2 Parameter Exploration

A dedicated parameter exploration module has been incorporated into the simulator to investigate the influence of different physical quantities on the resulting disk structure.

The current implementation supports systematic variation of parameters including

- Stellar mass.
- Disk mass.
- Characteristic radius.
- Surface density exponent.
- Temperature exponent.
- Reference temperature.
- Inner and outer disk radii.

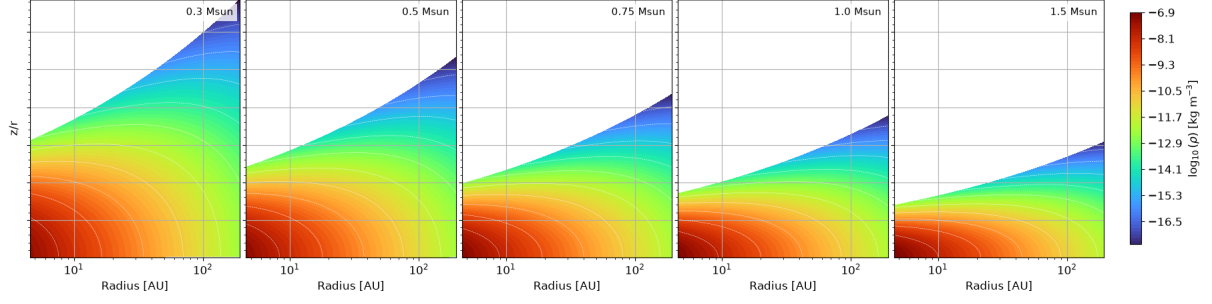


Figure 5.2: Effect of Varying stellar Mass on the Model for TW Hya, It can be seen that the increasing gravitational effect of the star flattens the disk.

Each parameter may be varied independently while maintaining identical initial conditions for all remaining quantities. This approach allows the physical influence of individual parameters to be investigated in a controlled manner.

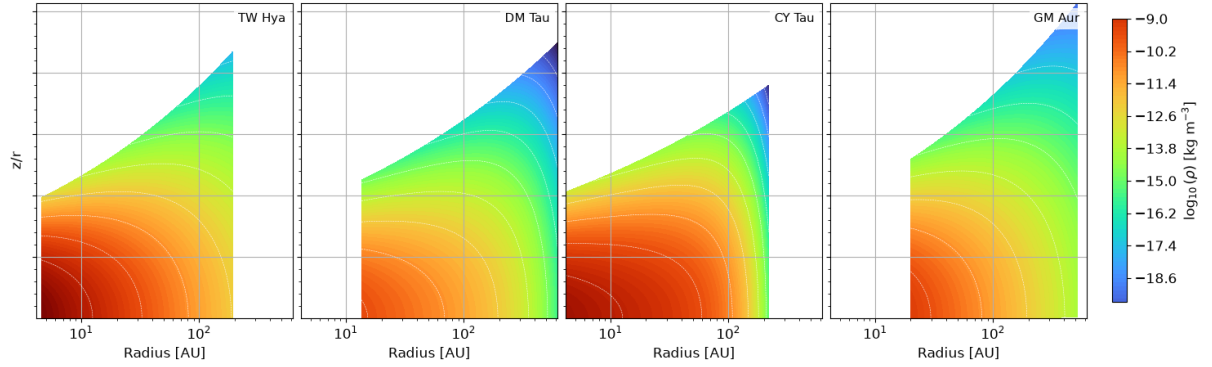


Figure 5.3: Models for different disks

5.3 Repository and Supplementary Material

Owing to the modular nature of the project, the simulator is capable of generating a large number of plots and comparative studies. Including every generated figure within the report would considerably increase its length while providing limited additional insight.

Instead, the complete implementation, together with all generated visualizations, parameter exploration studies, source code, and Jupyter notebooks, is provided through the accompanying GitHub repository. The repository serves as the primary source for reproducing the computational experiments and examining the simulator outputs in greater detail.

The technical report therefore focuses on the underlying physical model, mathematical formulation, and implementation methodology, while the repository contains the complete set of generated scientific results.

Chapter 6

Conclusion

In this work, a modular computational framework has been developed for modelling the structure of passive protoplanetary disks using simplified astrophysical assumptions. The simulator combines established analytical models describing the radial and vertical structure of the disk into a unified numerical pipeline capable of computing the fundamental physical properties of a star–disk system.

The current implementation successfully models the surface density, temperature distribution, pressure scale height, Keplerian rotation, and two-dimensional gas density structure. A modular software architecture has been adopted to ensure that additional physical processes such as planet–disk interactions, dust evolution, radiative transfer, and time-dependent disk evolution can be incorporated in future versions with minimal restructuring.

The project provides an educational as well as research-oriented platform for studying the influence of fundamental physical parameters on the morphology of protoplanetary disks. Further development will focus on extending the physical realism of the simulator while preserving its modular and computationally efficient design.

7 Repository

The complete source code, Jupyter notebooks, generated visualizations, and supplementary material associated with this project are available in the accompanying GitHub repository.

GitHub Repository:

https://github.com/Vedant-JNV/protoplanetary_disc

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Thank You

Thank you for taking the time to review our work.